

Predicting Customer Purchase Behavior Using Modeling Techniques

MIS 4560/5560 Project

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Project Description

- Focuses on predicting customer purchase behavior for an online retail store
- Transactional and behavioral data from a Kaggle dataset "eCommerce behavior from multi-category store"

Business Problem: Online retailers want to increase the rate of visitors becoming buyers

Project Goals

- Predict likelihood of purchase
- Identify key drivers of behavior
- Compare multiple model performance to determine most efficient approach for targeting marketing efforts

Dataset Overview



Data from Oct 2019 to Apr 2020



Includes user sessions and product behavior

- User session – relation between product and user
 - Visits, product views, session duration, product category, etc.

Target & Variables

- Target variable: **purchase (1 = yes, 0 = no)**
- Example predictors:
 - price
 - product_id
 - view_count
 - cart_count
 - session_duration

Data Preprocessing

- Binary conversion of purchase
- Group by product and session
- Aggregate columns products viewed, event count, duration, time of day bin, weekday, etc...
- No missing values
- 60/40 split

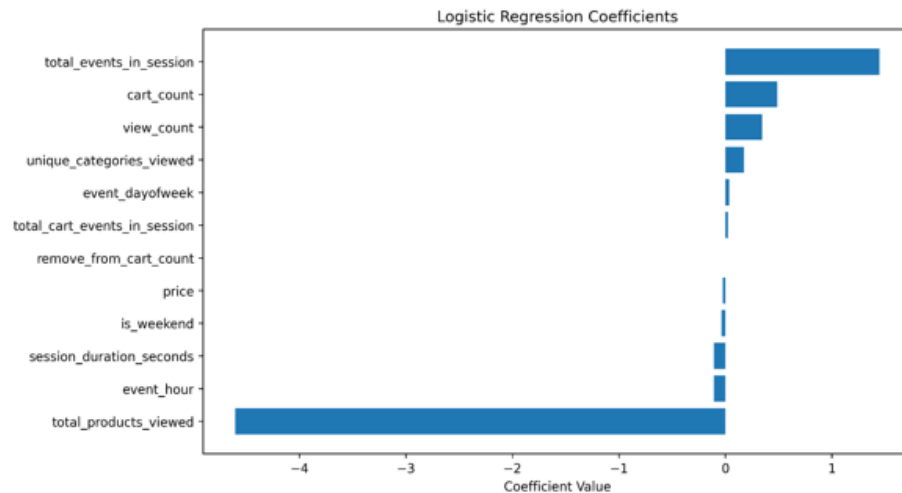
Models Used

Linear, Logistic, Lasso, Decision Tree, Random Forest, Neural Net

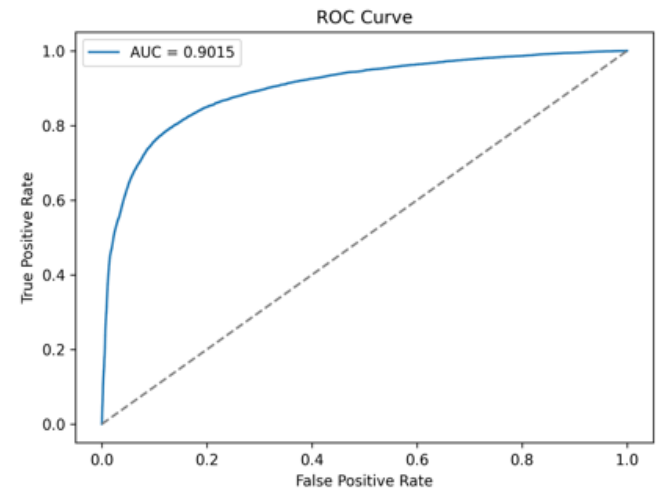
Linear Regression

- Used as a baseline model due to its simplicity and ease of interpretation
- As price increases, the probability of purchases slightly decreases
- $\text{Purchase} = 0.03690 - 8.48 \times 10^{-10} * (\text{product_id}) - 5.53 \times 10^{-6} * (\text{price})$
- F-value = 551.56
- P-value < 0.0001
- This indicates that the model is statistically significant
- $R^2 = 0.0043$
- This indicates that the model can only explain a very small portion of the variation in purchase behavior

Logistic Regression



	Pred 0	Pred 1
Actual 0	389020	1055
Actual 1	8726	1199



Metric	Value
Accuracy	0.975548
ROCAUC	0.901513

The logistic regression model provides a clear picture of e-commerce behavior. The model results in a high accuracy(97.5%) and strong ROC AUC(.9). It shows classic e-commerce behavior where long sessions with a lot of products_viewed (endless browsing) are far less likely to purchase with a co-efficient of -4.6 while event_count (adding to cart, viewing, etc) has the opposite effect with a co-efficient of 1.44.

Decision Tree

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Confusion Matrix:  
[[100443    0]  
 [  2683    0]]  
Accuracy: 0.9739832825863507  
  
Feature Importance:  
price : 0.147838175364426  
product_id : 0.8521618246355739
```

- A decision tree model was used to predict whether a customer would make a purchase based on price and product ID.”
- It achieved a high accuracy of about 97.4%, but when we looked closer at the confusion matrix, we found that the model predicted every case as ‘no purchase.
- The data is misleading because the model failed to identify any actual buyers.”

Random Forest

The random forest model improved predictive accuracy and reduced variance by aggregating multiple trees.

	feature	importance	std
2	remove_from_cart_count	0.000000	0.000000
11	is_weekend	0.006798	0.001687
5	unique_categories_viewed	0.019619	0.003886
10	event_dayofweek	0.037087	0.003972
6	total_cart_events_in_session	0.047575	0.036785
9	event_hour	0.063642	0.005650
4	total_products_viewed	0.082645	0.025428
7	price	0.101999	0.007754
3	total_events_in_session	0.126405	0.026972
1	cart_count	0.134654	0.065489
0	view_count	0.165154	0.044865
8	session duration seconds	0.214422	0.044494

	Prediction	
Actual	0	1
0	100278	229
1	952	1667

Neutral Net

Model	Accuracy	Elapsed(sec)
Small NN	0.98619	88.97384130000137
Medium NN	0.986295	194.3005497000413
Large NN	0.985345	499.8412876999937

Small	Predicted 0	Predicted 1
Actual 0	387982	1960
Actual 1	3564	6494

Medium	Predicted 0	Predicted 1
Actual 0	388424	1518
Actual 1	3964	6094

Large	Predicted 0	Predicted 1
Actual 0	388099	1843
Actual 1	4019	6039

- The NN model performed well with the overall accuracy among all three around 98%. The small and medium NN performed the best at 98.62 and 98.63, but the large takes a dive in accuracy. This shows that once the model is detailed enough to capture the decision boundary, adding more layers or neurons only increases training time and risk of overfitting without boosting predictive power by inserting unnecessary noise.

Summary

- Preprocessing of data/quality of data is critical to the outcome of your analysis
 - Remove irrelevant variables, zeros, simplify data
- Important to choose the right independent variables
 - Identified through correlation checks and lasso regression
- Complex models perform better than linear models